



Sizhang Xu

Full-Stack Engineering · Real-Time Systems

+1 425-380-1254

kelvinxu824@gmail.com

Greater Seattle Area, WA

xuxu8421.github.io

EDUCATION

University of Washington

Expected Jun 2028

M.S. in Technology Innovation · Global Innovation Exchange (GIX) · In progress

Tsinghua University

Graduate studies · GIX dual-degree program

Lab affiliation: PI Lab, Department of Computer Science and Technology

Shanghai Jiao Tong University

Jun 2025

B.Eng. in Industrial Design · Rank 1/30 · National Scholarship

Semester exchange: University College London

INTERNSHIP EXPERIENCE

ByteDance | Seed

Feb - Jun 2026

LLM Data Product Intern | **LangGraph** Evaluation Agent

- **DataSeek post-evaluation analysis:** Led the development of DataSeek's AI evaluation-analysis module from concept to iterative releases, turning quantitative model-evaluation results into capability-level failure diagnoses and training-data recommendations. Owned PRDs, prompt architecture, and analysis pipeline design; developed and iterated the **Python/LangGraph** prototype.
- **Analysis pipeline design:** Designed an agent pipeline spanning reference reasoning → comparative error diagnosis → capability aggregation → training-data mapping. Used **best-of-N** reference sampling and LLM judging to establish a reasoning baseline; compared model and reference reasoning to identify question-level failures and capability labels, then aggregated fine-grained labels into mid-level capabilities for pretraining, SFT, and RL data plans. Designed a **four-quadrant analysis framework** (reference answer present/absent × scoring rubric present/absent) for objective and subjective evaluations.
- **Prompt system design:** Led the design of **30+** structured prompt templates across the pipeline, defining scenario-action capability naming and constraints against unsupported capability attribution. Addressed diluted long-tail failures and spurious capability labels with two-stage aggregation: first bucket by 18 predefined capability labels, then adapt grouping within each bucket to the number of failure causes, preserving long-tail diagnoses and supporting cross-dataset comparison.
- **Product design and iteration:** Independently wrote PRDs and led multiple review rounds. Led manual quality checks of **700+** reports, identified nearly 100 **P0/P1/P2** issues, and established an issue → requirement → fix tracking loop. Designed short-term frontend fallbacks alongside longer-term prompt-engineering changes to address recurring problems.

Sohu | Mint Lab

Oct 2025 - Jan 2026

AI Product Management Intern

- **Knowledge-graph content creation:** Built an AI-assisted creative-content tool from the ground up for creators facing information overload, fragmented ideas, and difficulty organizing material. Organized the experience around a cognitive reasoning chain and the creative workflow, turning divergent ideas into editable, traceable knowledge graphs and drafts.
- **Prompt architecture and optimization:** Led prompt design across association → structuring → linear overview → drafting. Addressed uncontrolled content expansion with **Root/Hyponym** constraints and chain-of-thought prompting, combined with self-ranking and voting to reduce hallucinations and inconsistent parent-child relationships.
- **Engineering implementation:** Independently developed Python end-to-end test scripts. Split title and description generation into sequential model calls to optimize the execution pipeline; designed an Outline-to-Draft generation flow for granular editing and traceability.
- **Evaluation system:** Built a **six-dimension** quality rubric with stage-specific weights and a four-level evaluation set based on recurring content topics. Selected model combinations under quality, latency, and cost constraints; used regression cases to quantitatively validate prompt iterations and check stability.

INTERNSHIP EXPERIENCE · CONTINUED

Tsinghua University | Future Laboratory

Jun - Sep 2024

Human-Computer Interaction Research Intern

- **System modeling and prototyping:** Built a wearable force-feedback prototype and modeled the relationship between motor input angle and output resistance. Implemented three **C#/Unity** modules for cutting fruit and vegetables, chopping a wooden target, and tearing paper.
- **Testing and iteration:** Integrated virtual contact events with physical resistance and iterated interaction timing. Evaluated the prototype with **eight participants**, using observations and signal traces to investigate timing mismatches and compensatory effort near the end of a cutting motion.

PUBLICATIONS

"In Sync with Me": A Robotic Body-Doubling Probe for ADHD Programmers' Deep Work

Chenyang Li, **Sizhang Xu**, Hancheng Ye, and Ziyu Huang

UbiComp Companion '26 (Poster), 2026 · Accepted / in press

[DOI](#)

SELECTED PROJECTS

Real-Time Sleep Research Platform

2026

Python · **FastAPI** · Vue/JavaScript · **WebSocket** · BLE

- Implemented **FastAPI** REST controls and **WebSocket** runtime snapshots for connection setup, live signal inspection, intervention status, and event timelines. Exposed disconnected devices, stale readings, and cooldown conditions as explicit interface states.
- Built a Python research platform connecting BLE sensors, audio inference, intervention control, and a Vue interface. Separated hardware access, control logic, and recorded-session review so researchers could configure and inspect an experiment in one application.
- Separated audio capture, pretrained **TensorFlow/YAMNet** inference, and BLE I/O with a **ring buffer**, an inference worker, and a dedicated **asyncio** event loop. Kept model inference out of the audio-capture callback and used configurable thresholds and state smoothing.
- Implemented a trigger, playback, observation, and cooldown **state machine**. Recorded timestamped events in **JSONL** and moved buffered waveform compression and **NPZ** block writes to a background thread for session review.

Alpha-TradingAgents | Multi-Agent Equity Research

Summer 2026

Python · **LangGraph** · akshare | github.com/xuxu8421/alpha-trading-agents

- Extended the open-source TradingAgents framework for A-share equity research, adding market, financial, news, and policy data adapters. Implemented **exponential-backoff** retries around data requests and connected the resulting inputs to the existing analyst workflow.
- Added custom policy-analysis and report-writer nodes to **LangGraph**. Combined analyst evidence with the framework's bull/bear debate and risk-review outputs to produce a synthesized research report. The extension reused the framework's existing orchestration.
- Defined a structured metadata interface between report synthesis and PDF rendering, separating research logic from presentation. Built the report renderer to populate summary cards from those fields and generated a **four-page** research-report sample.

CAMPUS LEADERSHIP

SJTU Student Union — Presidium Member

School of Design Student Union, SJTU — Executive President

Nanyang News Agency, SJTU — Presidium Member & Deputy Secretary-General

SKILLS

Languages: Python, JavaScript, C#, HTML/CSS

Engineering: **LangGraph**, **FastAPI**, Vue, REST APIs, **WebSocket**, **asyncio**, Git

Evaluation & product: LLM evaluation, prompt design, regression testing, requirements, Figma